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Efficacy of essential oils combination on performance, ileal bacterial counts, intestinal histology and immunocompetence of laying hens fed alternative lipid sources

Asma Mousavi[1] | Amir Hossein Mahdavi[1] | Ahmad Riasi[1] | Masoud Soltani-Ghombavani[2]

[1]Department of Animal Sciences, College of Agriculture, Isfahan University of Technology, Isfahan, Iran

[2]Institute of Veterinary, Animal and Biomedical Sciences, Massey University, Palmerston North, New Zealand

Correspondence
Amir Hossein Mahdavi, Department of Animal Sciences, College of Agriculture, Isfahan University of Technology, Isfahan 84156-83111, Iran.
Email: mahdavi@cc.iut.ac.ir

Abstract

This study was carried out to assess the effects of a blend of herbal essential oils (namely *Thymus vulgaris*, *Mentha piperita*, *Rosmarinus offisinalis* and *Anethum graveolens*) and different dietary lipid sources on the performance, ileal bacterial enumeration, intestinal histological alterations and immune responses in laying hens. For this purpose, a total of 150 laying hens were randomly allocated to six experimental treatments with five replicates of five birds each. Dietary treatments consisted of three levels of a mixture of essential oils (0, 100 and 200 mg/kg) and two sources of lipid (soybean oil and fish oil containing different ratios of n-6 to n-3 fatty acids) fed to the animals during an 80-days feeding trial. Findings indicated that dietary administration of fish oil not only increased significantly the spleen relative weight ($p < 0.01$) and the antibody titre against Newcastle virus ($p < 0.05$) but also led to reductions in liver relative weight ($p < 0.05$) and feed conversion ratio ($p < 0.05$). Moreover, the mixture of herbal essential oils brought about declines in hepatic relative weight, heterophile to lymphocyte ratio ($p < 0.05$) and intestinal pathogen populations ($p < 0.01$). Mention must also be made of the improvements it made in spleen weight ($p < 0.01$), antibody titres against SRBC ($p < 0.01$) and Newcastle virus ($p < 0.05$), villus height to crypt depth ratio ($p < 0.01$), goblet cell numbers ($p < 0.05$), lamina propria lymphatic follicle diameters ($p < 0.01$) and feed conversion ratio ($p = 0.06$). It may be claimed that the enhancements observed in the performance of laying hens fed fish oil and 200 mg/kg of the blend of essential oils could have potentially been associated with improved intestinal health indices as well as cellular and humoral immune responses.

KEYWORDS

essential oils blend, fish oil, immune responses, intestinal histology, laying hens, performance

1 | INTRODUCTION

Polyunsaturated fatty acids, especially the N-3 ones, have been frequently demonstrated to act noticeable roles in reducing blood cholesterol, modulating the immune system, preventing cardiovascular diseases and attenuating inflammatory responses in gastrointestinal malignancies (Siddiqui, Harvey, & Zalog, 2008; Yu, Liu, Zhang, Wei, & Yang, 2017). These favourable health impacts of N-3 fatty acids call for efforts to produce N-3 enriched eggs. This is especially beneficial because, in addition to the enhancements in the amino acid and protein contents, dietary manipulation can alter the fat soluble vitamin and fatty acid levels in eggs (Ayerza & Coates, 2001; Lewis, Seburg, & Flanagan, 2000). However, enriched eggs are reportedly prone to lipoperoxidation due to the presence of free radicals (Baucells, Crespo, Barroeta, Lopes, & Groshorn, 2000); as a consequence, it influences adversely not only the animal's performance but also some of its physiological health indices (Rahman Alizadeh, Mahdavi, Rahmani, & Jahanian, 2015). The remedy most commonly employed involves dietary supplementation with synthetic antioxidants such as butylated hydroxyanisole (BHA), butylated hydroxytoluene, (BHT) and/or tert-butylhydroquinone (TBHQ; Melluzzi, Sirri, Manfreda, Tallarico, & Franchini, 2000). Given the undesirable effects of excessive intake of synthetic antioxidants such as liver expansion or even carcinogenesis (Dapkevicius, Venskutonis, Van Beek, & Linssen, 1998), researchers have recently turned to dietary administration of natural antioxidants, particularly herbal essential oils, as alternatives to the synthetic ones. In this regard, a number of studies have reported that supplemental herbal antioxidants in diets can not only be transferred and deposited in the peripheral tissues to end up in eggs, but that they also serve useful purposes in internal organs. Thyme (*Thymus vulgaris*), dill *(Anethum graveolens)*, rosemary (*Rosmarinus officinalis*) and peppermint *(Mentha piperita)* are some of the medicinal plants with strong antioxidant activities that are currently in focus.

Dietary inclusion of thyme has been shown to increase serum antioxidant content and decrease the transfer of free radicals to the egg (Botsoglou, Yannakopoulos, Fletouris, Tserveni-Goussi, & Fortomaris, 1997). In addition, improved performance has been reported in broilers (Attia, Bakhashwain, & Bertu, 2017) and laying hens (Abdel-Wareth, Ismail, & Südekum, 2013) receiving the essential oil or leaves, respectively, of *T. vulgaris* L.

Not only are carvone and limonene present in dill as its major essential oil components, but it also contains dill apiole, trans-dihydrocarvone and α-phellandrene in discernible amounts (Radelescu, Popescu, & Ilies, 2010). Dill has been found to possess antibacterial, antioxidant and hepatoprotective properties (Abbasi Oshaghi et al., 2017).

Rosemary has the main phenolic constituents including carnosic acid, carnosol, rosmanol and epirosmanol (Richeimer, Bernart, King, Kent, & Bailey, 1996). Alagawany and Abd El-Hack (2015) showed that dietary supplementation of rosemary by up to 6 g/kg of laying hens was able to improve egg production, egg mass, serum lipid profile, superoxide dismutase activity and blood immunoglobulin M concentration.

Such varied and beneficial activities as anti-inflammatory, antimicrobial, antioxidant and also cytotoxicity effects have been attributed to peppermint (Farzaei, Azizi, Arji, & Rostami, 2015). Dietary administration of peppermint has been reported to decrease egg yolk malondialdehyde level in hens exposed to heat stress (Bahrami, Shariatmadari, & Karimi Torshizi, 2011).

While there are some published reports on the dietary administration of essential oils in N-3 fatty acid-enriched diets, the synergetic effects of essential oil blends have recently received attention because of their biological potentials for improving the health indices and productive performance of hens (Adaszyńska-Skwirzyńska & Szczerbińska, 2017; Rimini, Petracci, & Smith, 2014). Inspired by these studies, this study was carried out to investigate the effects of combined essential oils (*T. vulgaris*, *M. piperita*, *Rosmarinus offisinalis* and *A. graveolens*) on the productive performance, immunocompetence, jejunal health and ileal microbial enumeration in laying hens supplemented with different dietary lipid sources (containing different ratios of n-6 to n-3 fatty acids).

2 | MATERIALS AND METHODS

2.1 | Experimental design and dietary treatments

A total of 150 Hy-Line W36 Leghorn laying hens, 40 weeks of age, were randomly allocated to six experimental diets with five cage replicates of five birds each. Dietary treatments consisted of a 2 × 3 factorial arrangement of treatments with three levels (0, 100 and 200 mg/kg of diet) of a blend of essential oils extracted from thyme, rosemary, dill, and peppermint and two sources of dietary lipid (soybean oil and fish oil). The experiment consisted of 10 days of adaptation followed by 70 days of main recording period. The birds were caged in a windowless poultry house provided with a light regime of 16L: 8D with free access to water and diets.

2.2 | Chemical analysis and measurements

The fatty acid contents of kilka fish oil, soybean oil and the experimental feeds were determined using chloroform: methanol (2:1, vol/vol) via gas chromatography (AOAC, 969.33) according to the method of Folch, Lee, and Sloane-Stanley (1957). Methanolic HCl, as the derivatizing agent, was used to derive fatty acid methyl esters from the total lipid extract. In brief, approximately 2 ml of the extracted lipid was evaporated to dryness under nitrogen at 40°C while 1 ml of anhydrous 3 N methanolic HCl was added. The aliquots were then stored in water at 60°C for 40 min, cooled to room temperature and extracted with hexane. Then, an Agilent 6890 gas chromatograph equipped with an autosampler and a flame ionization detector was employed to analyse fatty acid methyl esters before being subjected to a BPX-70 silica capillary column with a film thickness of 120 m × 0.25 mm × 0.2 μm, as reported by Rahman Alizadeh et al. (2015). Moreover, the chemical composition of basal ingredients including crude protein, ether extract, crude fibre, calcium and phosphorous (AOAC, 2006) was investigated. The exact values of the ingredients were used to formulate dietary treatments based on nutritional requirements of each bird strain (Table 1). The gas chromatography–mass spectrometry (GC–MS) was used to determine the major components of the essential oils mixture by the Agilent 7890 gas chromatographer (Agilent Technologies, Palo Alto, CA, USA) equipped with an HP-5MS 5% phenyl methyl siloxane capillary column, as described by Alizadeh, Alizadeh, Amari, and Zare (2013).

TABLE 1 Composition and nutrient content of basal diets (40 to 50 weeks of age)

Item	Soybean oil included diet	Fish oil included diet
Ingredients (g/kg)		
Corn	554.7	554.7
Soybean meal	219.4	219.4
Soybean oil	30.0	0.0
Fish oil	0.0	30.0
Wheat bran	62.6	62.6
Monocalcium phosphate	15.8	15.8
Calcium carbonate	105.6	105.6
Common salt	2.7	2.7
DL-Methionine	1.7	1.7
L-Lysine	0.2	0.2
Threonine	0.2	0.2
NaHCO3	2.0	2.0
Mineral premix[a]	2.5	2.5
Vitamin premix[b]	2.5	2.5
Nutrient composition		
Calculated analysis		
Metabolizable Energy (MJ/Kg)	11.30	11.30
Lysine (g/kg)	7.8	7.8
Methionine (g/kg)	4.3	4.3
Methionine + Cysteine (g/kg)	6.8	6.8
Threonine (g/kg)	5.9	5.9
Available phosphorous (g/kg)	4.6	4.6
Measured analysis		
Crude protein (g/kg)	152.5	152.5
Crud Fat (g/kg)	54.7	54.7
Calcium (g/kg)	43.5	43.5
Fatty acids (g/kg of total dietary fatty acids)		
Total saturated fatty acids	248.7	263.5
Total monounsaturated fatty acids	353.7	372.3
Total polyunsaturated fatty acids	397.6	364.2
Total n-6 fatty acids	382.5	262.0
Total n-3 fatty acids	15.1	101.8
n-6 to n-3 ratio	25.0	2.57

Note.[a]Vitamin premix provided the following per kilogram of diet: vitamin A (from vitamin A acetate), 9,800 IU; cholecalciferol, 2,100 IU; vitamin E (from dl-α-tocopheryl acetate), 22 IU; riboflavin, 4.4 mg; nicotinamide, 40 mg; calcium pantothenate, 35 mg; menadione, 1.50 mg; folic acid, 0.80 mg; thiamine, 3 mg; pyridoxine, 10 mg; biotin, 1 mg; choline chloride, 560 mg; and ethoxyquin, 125 mg. [b]Mineral premix provided the following per kilogram of diet: Mn, 65 mg; Zn, 55 mg; Fe, 50 mg; Cu, 8 mg; I [from $Ca(IO_3)_2 \cdot H_2O$], 1.8 mg; and Se, 0.30 mg.

2.3 | Weight measurement of internal organs

On days 70 of the trial, two birds per cage were randomly selected and sacrificed to assess their relative inner organ weights using a sensitive digital scale. The organs heart, liver, gallbladder, pancreas, spleen and oviduct were precisely weighed and expressed as percentage of live body weight.

2.4 | Immunological tests

On days 41 of the experiment, the hens received Newcastle vaccine injections. After one week, two randomly selected hens from each replicate were bled from their wing veins. Sera were separated by centrifugation (3,000 *g* for 10 min) and the Hemagglutination halter test was performed using the commercially available V-form ELISA plates to determine the antibody titres against the Newcastle disease virus (NDV) according to the procedure described in Taheri Gandomani, Mahdavi, Rahmani, Riasi, and Jahanian (2014). In addition, sheep red blood cells (SRBC) were washed in the phosphate buffer saline and diluted in this same buffer to a final dilution rate of 70 ml/L (vol/vol). On days 48 and 58 of the experiment, two hens per cage were injected (in the thigh muscle) with 1 mL of the 70 ml/L SRBC suspension. Heparinized blood samples were added on days 7 and 14 after the initial and second immunizations. Plasma samples were then prepared and kept at −20°C until further analysis. The hemagglutination test was performed according to the method described in Leshchinsky and Klasing (2001).

At the end of the experiment, two birds of each replicate (cage) were randomly selected and bled via wing vein. A Giemsa-stained slide was used to determine the differential leucocyte counts. The different leucocyte subpopulations were numbered and the heterophile to lymphocyte ratio was estimated as suggested by Stedman, Brown, BrooksJr, and Bounous (2001).

2.5 | Bacteriological examinations

The total ileal digesta content of two birds per replicate was carefully collected on days 70 of the trial from Meckel's diverticulum to the ileocecal junction. The ileal samples were completely mixed, poured into sterile tubes, stored in an ice-covered box, and instantly transferred to the laboratory. The ileal content was prepared to culture on Eosin Methylene Blue agar plates (Merck, Darmstadt, Germany), and the microflora populations were counted and confirmed using biochemical tests of indole, methyl red, Voges-Proskauer, and citrate reactions as previously explained by Mirbod, Mahdavi, Samie, and Mehri (2017).

2.6 | Measurement of jejunal morphological changes

The effects of dietary treatments on intestinal cell morphology were evaluated using the method described in Mirbod et al. (2017). In brief, two randomly selected birds from each pen were compassionately sacrificed at the end of the trial. Tissue samples were collected and a segment 1 cm in length was removed from the jejunal region anterior to Meckel's diverticulum for observation under a light microscope. The histological samples were fixed in 100 ml/L formaldehyde solution. The fixed samples were then embedded in paraffin. A microtome set was used to make transverse and longitudinal sections 5 μm in thickness before they were stained with haematoxylin-eosin and, finally, examined under a light microscope.

2.7 | Productive performance

Eggs were collected, counted, and immediately weighed every day (at 10:00 a.m.) throughout the experimental period. Egg mass was calculated based on egg numbers and weights. During the entire trial period, the performance traits including feed intake, egg production, egg weight, egg mass, and feed conversion ratio were determined at 35-days intervals.

2.8 | Statistical analysis

All the data were subjected to ANOVA using the GLM procedures of SAS software (SAS, 2001). The following model was assumed in the analysis of all the traits. $Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}$, where Y_{ijk} = observed value for a particular trait, μ = overall mean, A_i = effect of the i^{th} source of dietary lipid, B_j = effect of the j^{th} level of essential oils mixture, AB_{ij} = the respective interaction of i^{th} and j^{th} levels of dietary lipid source and essential oils mixture, and e_{ijk} = random error associated with the ijk^{th} recording. The treatment means were analysed by LSD tests. The threshold of significance was set at $p < 0.05$ and tendency towards statistical significance were declared at $0.05 < p < 0.10$. Normality was tested using the Shapiro-Wilk test and the histological alterations were analysed according to the GLM procedure of SAS software.

3 | RESULTS AND DISCUSSION

3.1 | Relative weights of internal organs

As reported in Table 2, the relative weight of spleen showed a significant increase ($p < 0.01$) while that of the liver exhibited a significant reduction ($p < 0.05$) after dietary inclusion of fish oil; however, no effects were observed on the relative weights of heart, pancreas, or oviduct. This finding is in agreement with those of Wang, Field, and Sim (2000) and Sijben, Schrama, Parmentier, van der Poel, and Klasing (2001), who reported no effects of fish, sunflower, flaxseed, and palm oils on the relative weights of inner organs in laying hens. Gibson and Roberfroid (2008) showed that the relative weight of liver is markedly associated with the liver fat percentage. Further support of our findings comes from Calder (1998) who reported reductions in spleen and thymus weights as a result of deficiency in, or lack of, essential fatty acids. In addition, it has been claimed that the marked increase observed in spleen weight induced by omega-3 fatty acids might have been the result of changes in fatty acid profiles which led to an increase in lipid storage in this organ for highly biological functions (Wang et al., 2000).

TABLE 2 Effects of different levels of the essential oils mixture on the relative weights of internal organs (g/kg live body weight) in laying hens fed different dietary lipid sources

Treatment		Oviduct	Gallbladder	Heart	Pancreas	Spleen	Liver
Lipid source							
Soybean oil		3.62	0.076	0.43	0.22	0.09^b	2.24^a
Fish oil		3.93	0.073	0.45	0.20	0.11^a	2.01^b
EOM (mg/kg)							
0.0		3.80	0.078	0.46	0.21	0.08^b	2.24^a
100		4.00	0.069	0.41	0.20	0.10^a	2.13^{ab}
200		3.52	0.075	0.45	0.21	0.11^a	2.01^b
Lipid source × EOM							
Soybean oil	0.0	3.40	0.063	0.48	0.22	0.06	2.30^a
Soybean oil	100	4.20	0.081	0.39	0.23	0.10	2.39^a
Soybean oil	200	3.25	0.083	0.42	0.21	0.09	2.04^b
Fish oil	0.0	4.19	0.094	0.45	0.20	0.09	2.19^b
Fish oil	100	3.81	0.056	0.43	0.18	0.11	1.86^c
Fish oil	200	3.79	0.068	0.48	0.22	0.11	1.97^c
p-value							
Lipid source		0.31	0.89	0.39	0.11	0.002	0.02
EOM		0.43	0.77	0.17	0.75	0.001	0.03
Contrasts		0.26	0.21	0.29	0.21	0.31	0.03
SEM		0.26	0.01	0.02	0.01	0.01	0.07

Note. Means with no common superscript letters are significantly ($p < 0.05$) different.
EOM: Essential oils mixture.

Inclusion of incremental levels of blended essential oils was observed to result in a significant ($p < 0.01$) reduction in the relative weight of liver but a remarkable ($p < 0.05$) increase in that of spleen (Table 2). Although Chowdhury, Chowdhury, and Smith (2002) found no differences in the relative weights of internal organs as a result of dietary inclusion of curcumin, thyme, mint and *Salvia officinalis*, Al-Soltan and Gameel (2004) reported a significant relationship between increased spleen weight and dietary inclusion of 0.1% turmeric in broiler chicks. The interaction between the experimental factors in the present study was only significant for liver relative weight; in turn, the lowest hepatic relative weight was obtained when the birds received diets containing fish oil simultaneous with the blend of herbal essential oils.

3.2 | Antibody responses to NDV and SRBC

Although feeding fish oil was only able to increase the antibody titre against NDV, dietary inclusion of at least 100 mg/kg of the blend of essential oils gave rise to a remarkable improvement in humoral responses against both NDV and SRBC antigens (Table 3). Improvements in the immune system functions as a result of dietary inclusion of n-3 fatty acids have been reported to occur due to changes in cell membrane compounds and structure, modulation in membrane signals and mediators (Robinson, Clandinin, & Field, 2001), and regulation in gene expressions or eicosanoides synthesis involved in inflammation (Peterson et al., 1998).

The bioactive compounds of essential oils might have been responsible for the elevated antibody titres against the experimental antigens (Recoqillay, 2006). As reported previously, improved antibody titre might be due to the antioxidant properties of aromatic herbs (Najafi & Torki, 2010) and their effects on enhancing the proportions of systemic lymphocyte as an antibody producer. Our in vitro trial indicated that the mixture of essential oils exhibited more than 1.5-fold higher antioxidant activity to inhibit 2-2-diphenyl-1-picric hydrazyl (DPPH) free radical (87.27 vs. 57.84 per cent) and 2.4-fold higher potential to reduce ferric ion (0.87 vs. 0.36 optical density) than BHT (data not shown). The GC–MS analysis showed that the major compounds detected in the mixture of essential oils were menthol (127.3 mg/g), γ-terpinen (109.2 mg/g), thymol (97.8 mg/g), carvone (87.5 mg/g), menthone (75.1 mg/g), carvacrol (62.7 mg/g), α-pinene (62.2 mg/g), 1,8 cineol (47.4 mg/g), camphor (39.7 mg/g), and *p* -cymene (32.4 mg/g), whose notable antioxidant and biological properties have been frequently confirmed. In agreement with our findings, Faramarzi et al. (2013) reported that aloe vera and yarrow feeds were capable of increasing antibody production against the Newcastle disease virus.

The interaction between the essential oils blend and dietary lipid sources were not significant with respect to humoral immune

TABLE 3 Effects of different levels of the essential oils mixture on the antibody titres against sheep red blood cell (SRBC) and Newcastle disease virus (Log_2) in laying hens fed different dietary lipid sources

			Sheep red blood cells	
Treatment		Newcastle disease virus	Primary response	Secondary response
Lipid source				
Soybean oil		7.27[b]	5.13	6.08
Fish oil		8.39[a]	5.63	6.00
EOM (mg/kg)				
0.0		6.84[b]	4.19[b]	4.50[b]
100		7.67[ab]	5.56[a]	6.75[a]
200		8.99[a]	6.38[a]	6.88[a]
Lipid source × EOM				
Soybean oil	0.0	6.28	4.38	4.50
Soybean oil	100	6.93	5.13	5.13
Soybean oil	200	8.60	5.88	5.88
Fish oil	0.0	7.40	4.00	4.50
Fish oil	100	8.41	6.00	7.00
Fish oil	200	9.37	6.88	5.50
p-value				
Lipid source		0.04	0.18	0.74
EOM		0.02	0.01	<0.01
Contrasts		0.07	0.25	0.15
SEM		0.31	0.31	0.42

Notes. Means with no common superscript letters are significantly ($p < 0.05$) different.
EOM: Essential oils mixture.

responses as evidenced by the enhanced antibody titre against NDV and/ or SRBC after dietary inclusion of essential oils in each of the dietary lipid sources.

3.3 | Differential circulating leucocyte counts

As shown in Table 4, the differential leucocyte counts and the heterophile to lymphocyte ratio remained unaffected by supplementation of fish or soybean oils. However, contrary to our findings, Yaqoob and Calder (1995) reported the suppression of lymphocyte proliferations in intra-tissue blood, spleen, thymus, and lymphoid nodules in rats fed diets enriched with excessive amounts of EPA and DHA. Using 200 mg/kg of dietary essential oils mixture was observed to reduce heterophils counts ($p < 0.05$), enhance lymphocyte counts ($p < 0.05$), but decrease the heterophil to lymphocyte ratio ($p < 0.05$) while no effects were observed on the enumeration of other blood leucocytes. The heterophil to lymphocyte ratio is an important index for evaluating the body immune status as its increase is associated not only with declining immunity and protection against pathogenic agents but also with the presence of physiological stresses (Sturkie, 1995). Hence, the reduction observed in this ratio in result of feeding 200 mg/kg of the blend in our study might be ascribed to the modulator and anti-inflammatory effects of essential oils. Consistent with these findings, it has been demonstrated that dietary consumption of peppermint (Awaad et al., 2010), aloe vera (Memeroele, 2011), or black seed (Soliman, Gahzalah, EL-Samra, Atta, & Abdo, 1999) has beneficial effects on the immunological activities of macrophages, proliferation of lymphocyte, and serum concentration of nitric oxide as well as humoral and cellular immune responses. In this regard, several studies have attributed the positive immunological effects of essential oils to their antioxidant properties. Because of their high biological activities, lymphocytes are highly susceptible to oxidative stress. It seems that oxidative stresses induce the destruction of cells involved in immunity, which results in the suppression of the immune system through prostaglandin E2 production (Bendich, Gabriel, & Machlin, 1986). Furthermore, we found interesting relationships between humoral immune responses and relative weights of some of the internal organs considered. Increased relative weight of the spleen, where lymphocyte maturation takes place, could likely be a reason for the enhanced lymphocyte counts in birds fed the essential oils blend. It, therefore, seems reasonable to hypothesize that the relative rise in lymphocyte counts following high levels of essential oils supplementation is due to the increases observed in the relative weights of secondary lymphoid organs like the spleen and also the protection of this organ against free radicals. In general, although feeding fish oil had no effect on white blood cell counts, the essential oils mixture administered at 200 mg/kg was found to induce a decline in the heterophil to lymphocyte ratio, as a physiological stress index.

TABLE 4 Effects of different levels of the essential oils mixture on differential circulating leucocyte counts in laying hens fed different dietary lipid sources

Treatment		Heterophile (%)	Lymphocyte (%)	Monocyte (%)	Eosinophile (%)	Basophile (%)	H:L ratio
Lipid source							
Soybean oil		31.91	65.75	1.37	0.79	0.16	0.49
Fish oil		34.09	63.54	1.45	0.81	0.09	0.54
EOM (mg/kg)							
0.0		35.31[a]	63.62[b]	1.50	0.81	0.12	0.57[a]
100		33.93[a]	62.50[b]	1.37	0.62	0.18	0.54[a]
200		29.14[b]	68.42[a]	1.35	1.00	0.07	0.42[b]
Lipid source × EOM							
Soybean oil	0.0	32.75	64.75	1.50	0.87	0.12	0.53
Soybean oil	100	33.62	64.00	1.37	0.62	0.37	0.51
Soybean oil	200	29.37	68.50	1.25	0.87	0.10	0.43
Fish oil	0.0	35.12	62.50	1.50	0.75	0.12	0.61
Fish oil	100	38.00	61.00	1.37	0.62	0.09	0.57
Fish oil	200	28.83	68.33	1.50	1.16	0.16	0.42
p-value							
Lipid source		0.35	0.31	0.69	0.74	0.54	0.35
EOM		0.03	0.04	0.84	0.18	0.75	0.02
Contrasts		0.68	0.79	0.85	0.59	0.16	0.71
SEM		1.18	1.06	0.24	0.24	0.13	0.04

Note. Means with no common superscript letters are significantly ($p < 0.05$) different.
EOM: Essential oils mixture; H:L ratio: Heterophile to lymphocyte ratio.

3.4 | Ileal microbial counts

The populations of *Escherichia coli* and *Salmonella* in the ileal content remained unaffected by administration of different dietary sources of oil (Table 5). In contrast, dietary inclusion of incremental levels of blends of essential oils led to a remarkable decline in *Escherichia coli* enumeration ($p < 0.01$) and tended to decline *Salmonella* colonies ($p = 0.06$). The interaction between treatments had no significant effect on intestinal bacterial counts. Ideal populations of intestinal bacterial flora are crucial for maintaining optimal performance, and any changes in the normal flora through dietary interventions or environmental factors might result in serious damages to the host's body functioning (Langhout, 2000). Hence, essential oils seem to have been capable of changing the intestinal microflora populations to give rise to the presence of ideal microbial flora for optimal growth (Brenes & Roura, 2010). Antimicrobial properties of individual or mixtures of essential oils have been known for many centuries, and several studies have been conducted to investigate their effects on the functions of micro-organisms (Brenes & Roura, 2010). Changes in intestinal microbial flora after consumption of aromatic plants have been generally attributed to their antimicrobial characteristics (Gulluce et al., 2007). Researchers have indicated that penetration of active constituents present in the extracts of medicinal plants into the pathogenic bacterial membranes causes changes in the transport of H+ and K+ ions by carrier proteins which, in turn, leads to impaired enzymes involved in synthetic reactions and, consequently, those taking place in the cell (Bolukbasi & Erhan, 2007). Helander et al. (1998) reported that thymol and carvacrol contained in thyme destroy *Escherichia coli* membranes. Moreover, it has been suggested that the bioactive compounds in essential oils are capable of perforating the bacterial membranes resulting in the leakage of vital intracellular substances (such as ribose and sodium glutamate), defects in bacterial enzyme systems, and thereby, impairment of the electron transfer, nutrient absorption, nucleic acid synthesis, and ATPase enzyme activity (Helander et al., 1998). Similar to that, cinnamaldehyde present in cinnamon has been found to play a pharmacological role in balancing intestinal microflora by selective inhibition of pathogenic bacteria resulting in improved production traits of broiler chicks (Ciftci, Simsck, Yuce, Yilmaz, & Dalkilic, 2010). Thus, although the populations of experimental ileal bacteria were not affected by different dietary lipid sources, dietary administration of the essential oils mixture

TABLE 5 Effects of different levels of the essential oils mixture on jejunal histological changes and ileal *Escherichia coli* and *Salmonella* enumerations ($\log_{10}$ CFU/g) in laying hens fed different dietary lipid sources

Treatment		Villus height (µm)	Crypt depth (µm)	VH:CD	LLF number	LLF diameter (µm)	GCN	*Escherichia coli*	*Salmonella*
Lipid source									
Soybean oil		1072.60	57.75	17.41	1.05+[2]	31.87	3.46+	5.67	3.68
Fish oil		994.50	55.81	16.50	1.07+	35.50	2.84+	5.53	3.62
EOM (mg/kg)									
0.0		903.70^b	59.00	14.00^b	1.00+	26.95^b	1.87^b+	6.64^a	4.01
100		1069.50^a	52.89	21.06^a	1.16+	35.93^a	3.07^a+	5.22^b	3.91
200		1158.89^a	58.45	19.75^a	1.00+	38.17^a	4.52^a+	4.93^b	3.02
Lipid source × EOM									
Soybean oil	0.0	997.33	58.00	15.87	1.00+	25.66	2.18+	6.70	4.16
Soybean oil	100	1108.25	52.99	23.74	1.16+	33.36	3.52+	5.41	3.73
Soybean oil	200	1134.40	61.38	16.20	1.00+	36.57	4.68+	4.90	3.14
Fish oil	0.0	763.00	59.12	11.19	1.00+	28.23	1.51+	6.58	3.87
Fish oil	100	1030.75	52.79	17.38	1.16+	38.50	2.63+	5.03	4.09
Fish oil	200	1189.50	55.51	22.94	1.00+	39.87	4.37+	4.97	2.91
p-value									
Lipid source		0.10	0.53	0.45	0.98	0.08	0.35	0.55	0.54
EOM		0.008	0.22	0.005	0.15	0.002	0.04	0.001	0.06
Contrasts		0.09	0.67	0.06	0.99	0.86	0.11	0.49	0.65
SEM		36.34	9.33	1.29	0.23	3.14	0.16	0.42	0.19

Note. Means with no common superscript letters are significantly ($p < 0.05$) different.
VH:CD: villus height to crypt depth ratio; LLF: lamina propria lymphatic follicles; GCN: goblet cell numbers; EOM: Essential oils mixture.
Number of +'s indicates severity of the histological changes.

was able to suppress the intestinal *Escherichia coli* and *Salmonella* enumerations.

3.5 | Intestinal histological changes

Administration of fish oil led to a marginal decrease in villus height ($p = 0.10$) and an increase in the diameter of lamina propria lymphatic follicles ($p = 0.08$), while it had no effect on other jejunal histological alterations (Table 5). On the other hand, dietary supplementation of at least 100 mg/kg of the essential oils mixture enhanced villus height ($p < 0.01$), villus height to crypt depth ratio ($p < 0.01$), and lamina propria lymphatic follicles extent ($p < 0.01$). The increased villus height to crypt depth ratio is probably due to a rise in villus height resulting from the bioactive compounds present in essential oils. This claim is based on the fact that these constituents are able to induce changes in the gastrointestinal tract histology, chemical composition of digestive juices, secretion of digestive enzymes, tissues related to gastrointestinal tract, and they, thereby, enhance the digestibility of the diet, leading to improved growth and performance. Supplementation of diet with essential oils has been demonstrated to be effective in reducing intestinal ammonia content in birds (Hong, Steiner, Aufy, & Lien, 2012), which is mostly attributed to the antimicrobial effects of essential oils. In agreement with our findings, Michiels et al. (2010) showed a rise in small intestinal villus height in piglets fed a diet enriched with carvacrol as the active compound of oregano. Hong et al. (2012) found that broilers supplemented with the essential oil derived from oregano had longer intestinal villi, as compared to their control and antibiotic treatments. Furthermore, diets enriched with flavonoids and essential oils reportedly led to increased intestinal villus height in weaning animals (Sehm, Lindermayer, Dummer, Treutter, & Pfaffl, 2007). Inclusion of thyme in broiler diets has been shown to be associated with a rise in intestinal villus height and, thereby, with increased intestinal absorption due to enlarged absorptive surface area (Garcia, Catala Gregori, Hemandez, Megras, & Madrid, 2006). As the major impact of plant extracts is to improve small intestinal microflora, this effect might be suggested as a factor changing intestinal morphology. It could be postulated that the reduction in small intestinal pathogenic bacteria resulting from the essential oils blend used in the present study led to the increased villi height, which improved the feed conversion ratio as a consequence (See below).

Crypt depth and lymphatic follicle numbers were not influenced by inclusion of the essential oils mixture. Crypt consists of several specialized absorptive, goblet, and generative cells which are responsible for both producing mucus and replacing aged cells. Crypt is considered as a villus-producing factory whose mitogenic activity and depth enhance when the need for cellular replacement increases (Yaqoob, Pala, Cortina-Borja, Newsholme, & Calder, 2000). It might then be postulated that the non-increasing crypt depth observed in the current study provides evidence for the fact that there was no need for high rates of cellular replacement.

Goblet cells are responsible for the production of mucus which has a protective function in the intestine. Therefore, the marginal increase in goblet cell numbers ($p < 0.05$) due to the inclusion of the essential oils mixture used in the present study could be attributed to the stimulating properties of its effective constituents. These compounds stimulate goblet cells proliferation, which induces the production of sufficient amounts of mucin to provide increased protection of the intestinal epithelium against pathogens; hence, the improved functional traits.

In conclusion, different dietary lipid sources were found to have no significant effects on the jejunal histological indices, while administration of at least 100 mg/kg of the essential oils blend improved not only the villus height to crypt depth ratio but also the lamina propria lymphatic follicle extent and goblet cell numbers, as intestinal health indices.

3.6 | Productive performance traits

Although fish oil administration lessened feed intake during the first 35 days of the experiment ($p < 0.05$), this trait remained unaffected during the second 35 days and the whole trial period (Table 6). Although fish oil had no effect on egg weight, it was observed to improve egg production ($p = 0.05$), egg mass ($p = 0.06$) and, subsequently, feed conversion ratio ($p < 0.05$) over the whole of experimental period.

Baucells et al. (2000) and Yuming, Chen, Xia, and Yuan (2004) reported no effects of different dietary lipid sources on feed intake; nevertheless, Gonzalez Esquerra and Leeson (2000) showed a decrease in feed consumption by laying hens fed diets enriched with a high level of fish oil (70 g/kg of diet). Furthermore, although Gonzalez Esquerra and Leeson (2000) and Yuming et al. (2004) indicated a reduction in egg weight as a result of including dietary fish oil, some experiments have shown that egg weight (Celebi & Macit, 2009), egg production percentage and egg mass (Grobas, Mendez, Laano, & Meteous, 2001), and feed conversion ratio (Guo, Chen, Xia, & Yuan, 2004) in laying hens were not affected by various sources of dietary lipid.

The essential oils blend used in the present study was observed to have no effects on feed intake or egg weight (Table 6). However, it induced improvements in egg production ($p < 0.05$) and feed conversion ratio ($p = 0.06$) during the second 35-day period. This is while some reports have indicated the positive effects of different levels or species of medicinal plants or their extracts on such productive traits as egg weight (Yalcin et al., 2009), egg production, egg mass (Bolukbasi, Urusan, Erhan, & Kiziltunc, 2010) and feed conversion ratio (Hong et al., 2012; Yalcin et al., 2009). Meanwhile, there also exist reports that showed no significant effects on these traits in laying hens. At last, the interactions among the experimental factors investigated in the present study did not affect performance traits.

Generally speaking, although dietary inclusion of fish oil lessened the feed conversion ratio over the whole experimental period, feeding incremental levels of the essential oils mixture improved egg production, egg mass and, subsequently, feed conversion ratio during the second experimental period. The improvements observed might be attributed to the beneficial effects of the essential oils mixture

TABLE 6 Effects of different levels of the essential oils mixture on the productive performance of laying hens fed different dietary lipid sources

		Egg production (%)			Egg weight (g)			Egg mass (g/ days per bird)			Feed intake (g)			Feed conversion ratio (g of feed: g of egg)		
Treatment		0–35	35–70	0–70	0–35	35–70	0–70	0–35	35–70	0–70	0–35	35–70	0–70	0–35	35–70	0–70
Lipid source																
Soybean oil		80.95	77.71	79.33	60.98	60.90	60.95	49.34	47.32	48.35	102.42[a]	97.20	99.81	2.08[a]	2.05	2.06[a]
Fish oil		84.14	81.26	82.69	61.89	60.85	61.46	52.07	47.82	50.03	99.00[b]	96.77	97.89	1.91[b]	2.03	1.96[b]
EOM (mg/kg)																
0.0		81.14	76.14[b]	78.64	62.22	61.07	61.78	50.47	46.50	48.58	102.62	97.55	100.09	2.05	2.10	2.07
100		83.57	80.00[ab]	81.78	60.77	60.52	60.66	50.78	48.93	50.99	99.25	96.08	97.66	1.95	2.05	1.99
200		82.92	82.32[a]	82.62	61.32	61.03	61.78	50.86	49.71	52.24	100.28	97.33	98.80	1.98	2.00	1.99
Lipid source × EOM																
Soybean oil	0.0	79.43	76.15	77.79	60.62	60.53	60.59	48.09	46.09	47.13	103.60	96.67	100.14	2.18	2.10	2.14
Soybean oil	100	81.71	77.28	79.50	60.89	61.17	61.04	49.75	45.73	47.75	102.28	96.74	99.52	2.05	2.12	2.08
Soybean oil	200	81.71	79.71	80.71	61.45	60.00	61.23	50.19	46.98	48.59	101.39	98.17	99.78	2.02	2.09	2.05
Fish oil	0.0	82.85	76.14	79.50	63.83	61.61	62.97	52.86	45.46	49.16	101.64	98.42	100.04	1.93	2.18	2.04
Fish oil	100	85.42	82.71	84.07	60.65	59.88	60.28	52.51	47.88	50.10	96.21	95.41	95.81	1.86	1.99	1.91
Fish oil	200	84.14	84.92	84.53	61.19	61.07	61.13	51.48	51.86	51.67	99.17	96.49	97.83	1.92	1.89	1.91
p-value																
Lipid Source		0.23	0.09	0.053	0.36	0.95	0.53	0.12	0.08	0.06	0.04	0.85	0.2	0.04	0.09	0.04
EOM		0.73	0.04	0.20	0.48	0.77	0.53	0.98	0.07	0.33	0.23	0.85	0.42	0.53	0.06	0.09
Contrasts		0.97	0.99	0.98	0.26	0.38	0.27	0.69	0.87	0.79	0.50	0.8	0.61	0.62	0.89	0.80
SEM		1.18	1.08	1.15	1.18	0.83	0.99	1.29	1.01	1.14	0.81	1.16	1.21	0.03	0.04	0.03

Note. Means with no common superscript are significantly ($p < 0.05$) different.
EOM: Essential oils mixture.

used that not only contributed to the inhibition of ileal pathogenic bacteria but also to the improvement of intestinal morphology, immunological functions and antioxidant indices.

ORCID

Amir Hossein Mahdavi http://orcid.org/0000-0001-5456-7625
Ahmad Riasi http://orcid.org/0000-0002-2119-2350

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